

## ASK

### **What is the problem you are trying to solve?**

The goal is to analyze smart device usage data to identify trends in how consumers use fitness devices. These insights will then be applied to a Bellabeat product to help improve the company's marketing strategy.

### **How can your insights drive business decisions?**

My insights can help Bellabeat improve its products, better understand customer needs, and develop marketing strategies that could attract more users.

### **Key Tasks**

Analyze smart device usage data to identify trends in consumer behavior and determine how these insights could be applied to Bellabeat customers and used to support Bellabeat's marketing strategy.

### **Key Stakeholders:**

Urška Sršen – Bellabeat cofounder and Chief Creative Officer  
Sando Mur – Bellabeat cofounder and member of the executive team  
Bellabeat marketing analytics team

## PREPARE

### **WHERE IS YOUR DATA LOCATED?**

The data is located on Kaggle in the FitBit Fitness Tracker Data dataset, made available by Mobius.

### **HOW IS THE DATA ORGANIZED?**

The dataset is organized into multiple CSV files containing different types of fitness and wellness data, including physical activity, steps, calories, heart rate, sleep, and weight. Most files use a unique user ID (Id) that can be used to connect data across datasets. The data is recorded at different levels of detail, including daily, hourly, minute-level, and second-level observations. Not all data types are available for every user.

### **ARE THERE ISSUES WITH BIAS OR CREDIBILITY IN THIS DATA? DOES YOUR DATA ROCCC?**

The dataset has some limitations that may affect its credibility. The original dataset description refers to 30 Fitbit users; however, the daily activity file used in this analysis contains 33 unique user IDs. Despite this difference, the sample is relatively small and may not represent the broader population of smart device users. The data was collected from users who voluntarily shared their fitness tracker data, which may also introduce selection bias. Therefore, the findings should be interpreted with caution.

## **HOW ARE YOU ADDRESSING LICENSING, PRIVACY, SECURITY AND ACCESSIBILITY?**

The dataset is publicly available on Kaggle under the CC0: Public Domain license. The Fitbit users consented to sharing their personal fitness tracker data. The dataset uses user IDs rather than names, which helps protect users' identities. The data is publicly accessible and can be used for analysis.

## **HOW DID YOU VERIFY THE DATA'S INTEGRITY?**

The data's integrity was verified by checking the number of records, unique users, date range, missing values, and duplicate records. The daily activity dataset contained 940 records from 33 unique users covering the period from April 12 to May 12, 2016. No duplicate records were found based on user ID and activity date.

## **HOW DOES IT HELP YOU ANSWER YOUR QUESTION?**

The dataset contains information about users' daily activity, including steps, activity levels, calories burned, and sleep. This data can be used to identify patterns in smart device usage and understand consumer activity and sleep behaviors that may help inform Bellabeat's marketing strategy.

## **ARE THERE ANY PROBLEMS WITH THE DATA?**

Yes. The dataset has several limitations. It represents a relatively small sample of Fitbit users and covers only about one month of activity. The data was voluntarily shared, which may introduce selection bias. Sleep data is also available for only a subset of the activity records. Therefore, the findings may not represent the broader population of smart device users and should be interpreted with caution.

## **PROCESS**

### **WHAT TOOLS ARE YOU CHOOSING AND WHY?**

I used Python with the pandas library for initial data inspection and cleaning, and BigQuery with SQL for data organization and analysis. Google Sheets was used to create the final visualizations.

### **HAVE YOU ENSURED YOUR DATA'S INTEGRITY?**

I used Python and pandas to check the integrity of both datasets. I checked their size, duplicate records, missing values, and data types using commands such as:

```
daily_activity.shape
```

```
daily_activity.duplicated().sum()
```

```
daily_activity.isnull().sum()
```

```
daily_activity.info()
```

```
sleep_day.shape
```

```
sleep_day.duplicated().sum()
```

```
sleep_day.isnull().sum()
```

```
sleep_day.info()
```

I found that daily\_activity contained 940 rows and 15 columns, with no duplicate or missing values. The ActivityDate column was stored as an object instead of a datetime.

The sleep\_day dataset contained 413 rows and 5 columns, with 3 duplicate records and no missing values. The SleepDay column was also stored as an object instead of a datetime.

The daily activity dataset contains 33 unique users, while the sleep dataset contains 24 unique users. This means that sleep data is not available for all users included in the daily activity dataset.

Both datasets cover the same period, from April 12, 2016 to May 12, 2016. This allows the activity and sleep data to be compared for matching dates.

I also checked the minimum and maximum values of key activity and sleep variables to identify potentially invalid values. I did not find any negative or clearly invalid values.

### **WHAT STEPS HAVE YOU TAKEN TO ENSURE THAT YOUR DATA IS CLEAN?**

I removed 3 duplicate records from the sleep dataset using drop\_duplicates(), leaving 410 rows and 5 columns. I also converted the ActivityDate and SleepDay columns from object to datetime format using pd.to\_datetime().

### **HOW CAN YOU VERIFY THAT YOUR DATA IS CLEAN AND READY TO ANALYZE?**

I verified that the data is clean by checking for duplicate records, missing values, incorrect data types, and invalid values. I also checked the number of unique users, the date ranges, and confirmed that each user has only one record per date. After cleaning the identified issues, I ran the checks again to confirm that the datasets are consistent and ready for analysis.

### **HAVE YOU DOCUMENTED YOUR CLEANING PROCESS SO YOU CAN REVIEW AND SHARE THOSE RESULTS?**

Yes. I documented each step of the data cleaning process, including the issues I identified, the changes I made, and the Python commands I used. This makes the cleaning process easy to review and reproduce.

## **ANALYZE**

### **How should you organize your data to perform analysis on it?**

The data was organized into separate tables in BigQuery based on the type of information. Daily activity data and sleep data were analyzed separately and then joined using the user ID and date to explore relationships between sleep and physical activity.

### **Has your data been properly formatted?**

The daily activity data was properly formatted in BigQuery, with appropriate data types for dates and numerical values. The SleepDay column was initially imported as a string and was converted to a date during the analysis to allow the sleep and activity tables to be joined correctly.

### **What surprises did you discover in the data?**

77 daily records (8.19% of the dataset) contained zero recorded steps. This may represent days with no activity, but it could also indicate that users did not wear or use their devices on those days.

Another interesting finding was that sleep duration had little relationship with daily activity. The correlation between sleep and steps was only -0.187, while the correlation between sleep and calories was -0.029.

### **What trends or relationships did you find in the data?**

Users spent substantially more time sedentary than being moderately or very active. On average, users recorded about 21 minutes of very active time and 14 minutes of fairly active time per day.

Users take an average of approximately 7,638 steps per day and travel an average distance of 5.49 units recorded by the dataset. Daily step counts vary considerably, ranging from 0 to 36,019 steps.

Daily activity varies by day of the week. Users recorded the highest average number of steps on Saturday (8,153) and Tuesday (8,125), while Sunday had the lowest average at 6,933 steps.

There is a moderate positive correlation (0.592) between daily steps and calories burned. Days with more steps tend to be associated with higher calorie expenditure.

Only 7 out of 33 users (about 21%) averaged at least 10,000 steps per day, while 26 users (about 79%) averaged fewer than 10,000 steps. This shows that most users in the dataset did not reach the 10,000-step threshold on average.

Users slept an average of approximately 419 minutes (about 7 hours) per night, while spending an average of 459 minutes (about 7 hours and 39 minutes) in bed. This represents a difference of approximately 40 minutes between time in bed and actual sleep time.

The analysis found no strong relationship between sleep duration and daily physical activity. The correlation between sleep duration and daily steps was weak and negative (-0.187), while the correlation between sleep duration and calories burned was close to zero (-0.029).

Users spent an average of 91.68% of their time in bed asleep, indicating a relatively small difference between time in bed and actual sleep time in this dataset.

## **How will these insights help answer your business questions?**

These insights help identify how consumers use fitness devices and highlight opportunities for Bellabeat to encourage greater daily activity. The findings could support marketing strategies focused on reducing sedentary time, encouraging regular movement, and helping users monitor both activity and sleep patterns.

## **SHARE**

### **Were you able to identify how consumers use smart fitness devices and what trends appear in their activity and sleep data?**

Yes. The analysis identified several patterns in how consumers use smart fitness devices. Users averaged approximately 7,638 steps per day and spent most of their recorded time sedentary, with relatively little time spent in moderate or intense activity. Activity levels varied across the week, and higher step counts were moderately associated with higher calorie expenditure. Users slept about seven hours per night on average, while sleep duration showed little relationship with daily steps or calories burned.

### **What story does your data tell?**

The data shows that users generally lead relatively sedentary lifestyles, with limited time spent in moderate or intense physical activity. Most users averaged fewer than 10,000 steps per day, although activity levels varied depending on the day of the week. Higher step counts were moderately associated with higher calorie expenditure. Users slept about seven hours per night on average, but sleep duration showed little relationship with daily physical activity.

### **How do your findings relate to your original question?**

The findings directly address the original business question by identifying key patterns in how consumers use fitness devices. The analysis shows high sedentary time, relatively low levels of moderate and intense activity, differences in activity across the week, and patterns in sleep behavior. These insights can help Bellabeat understand user behavior and identify opportunities to encourage more consistent daily activity through its products and marketing strategy.

### **Who is your audience? What is the best way to communicate with them?**

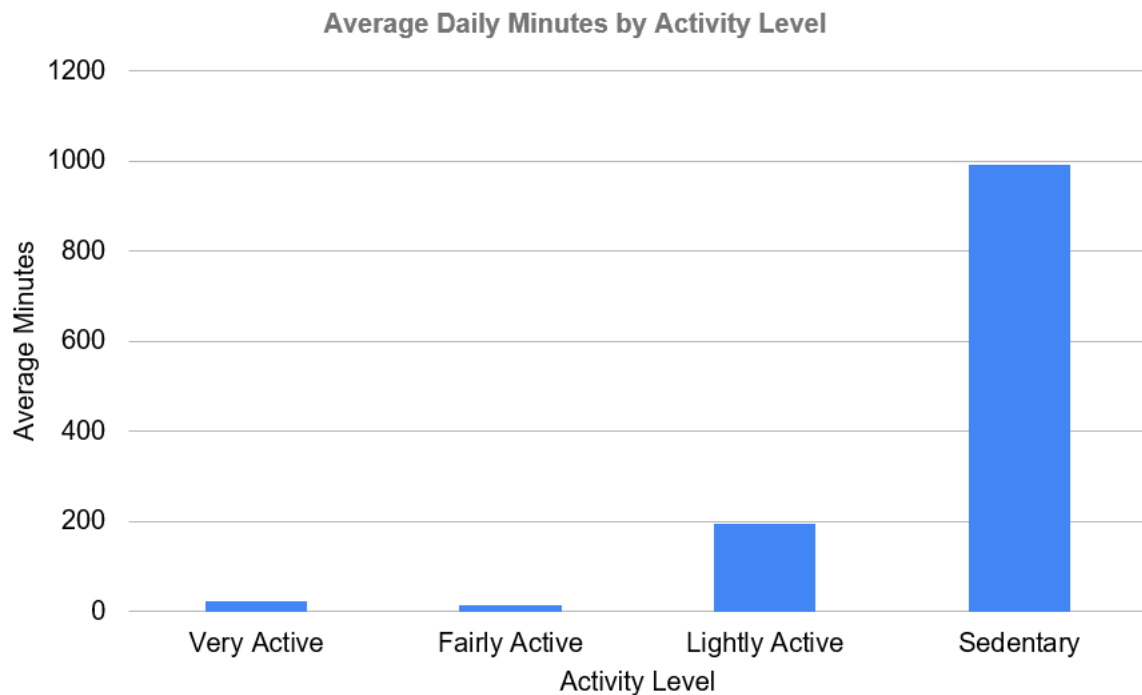
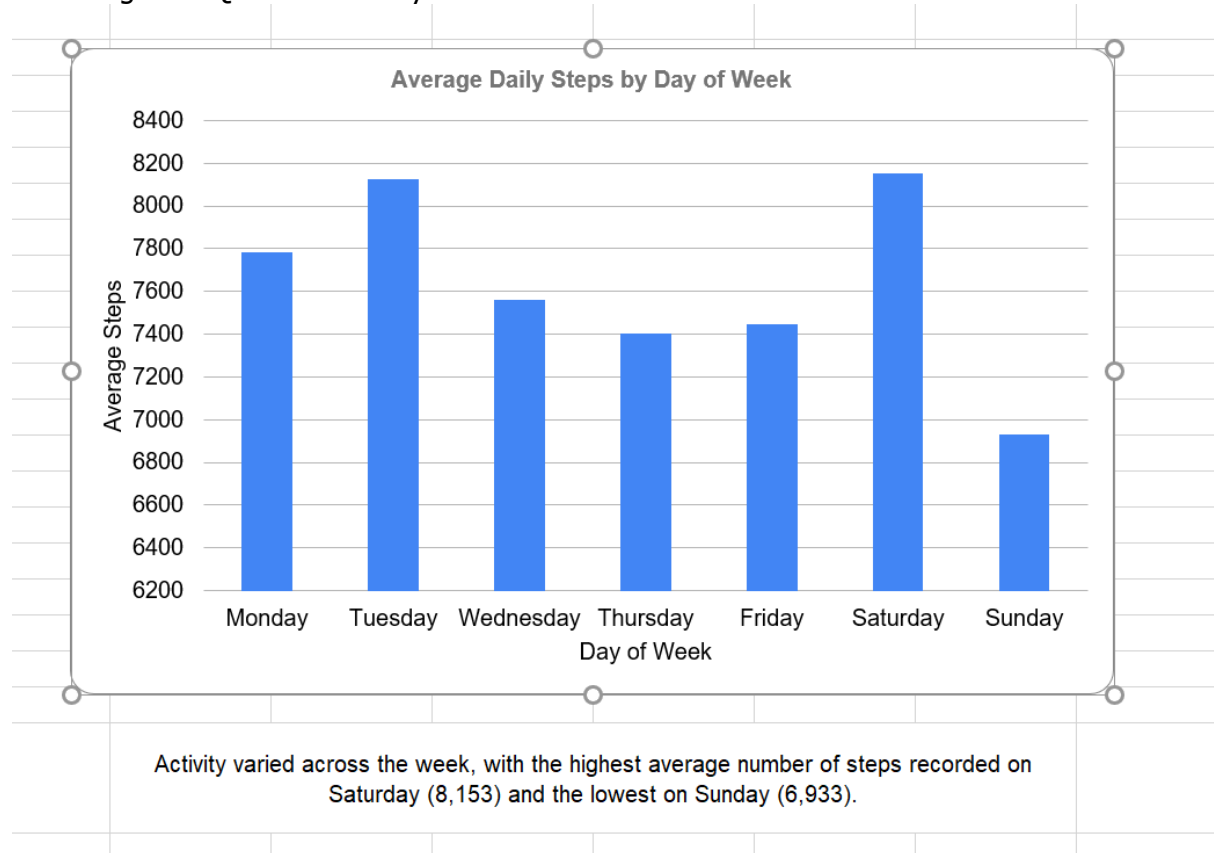
The primary audience includes Bellabeat cofounders Urška Sršen and Sando Mur, as well as the Bellabeat marketing analytics team. The findings should be communicated through clear and concise visualizations that highlight the most important trends and connect them to potential marketing opportunities.

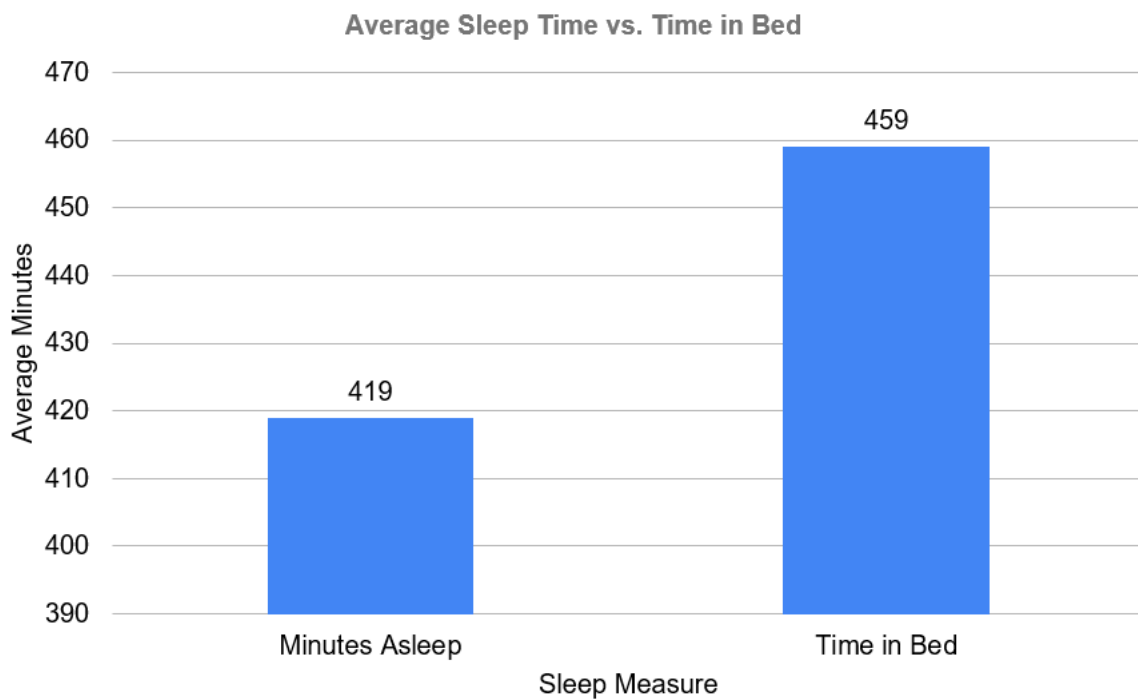
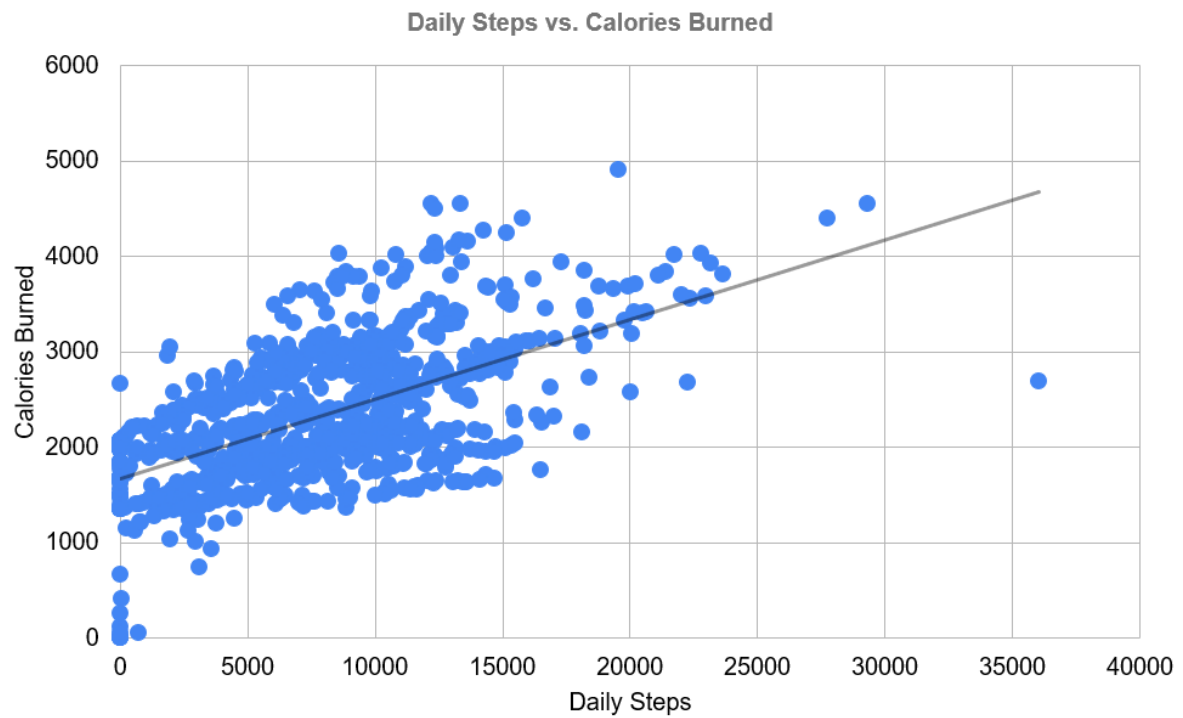
### **Can data visualization help you share your findings?**

Yes. Data visualizations can make the key trends easier to understand and compare, such as activity levels across the week, sedentary versus active time, and the relationship between steps and calories burned.

### Is your presentation accessible to your audience?

Yes. The presentation will use clear titles, simple charts, readable labels, and concise explanations so that the key findings can be understood without requiring technical knowledge of SQL or data analysis.





## Key Findings

The analysis identified several key patterns in smart device usage:

- Users averaged approximately 7,638 steps per day, with the highest average activity recorded on Saturday (8,153 steps) and the lowest on Sunday (6,933 steps).
- Users spent most of their recorded time sedentary, averaging approximately 991 sedentary minutes per day, compared with only 21 very active minutes and 14 fairly active minutes.
- Daily steps and calories burned showed a moderate positive correlation ( $r = 0.592$ ), meaning that days with more steps tended to be associated with higher calorie expenditure.
- Users slept approximately 419 minutes (about 7 hours) per night while spending approximately 459 minutes (about 7 hours and 39 minutes) in bed.
- Sleep duration showed no strong relationship with daily physical activity in the analyzed data.

## Act

### **What is your final conclusion based on your analysis?**

The analysis shows that users spend a large amount of their recorded time sedentary and relatively little time in moderate or intense activity. Daily activity varies across the week, and higher step counts are moderately associated with higher calorie expenditure. Users sleep approximately seven hours per night on average, while sleep duration shows little relationship with daily activity. Overall, the findings suggest that increasing regular daily movement is a key opportunity for Bellabeat to support its users.

### **How could your team and business apply your insights?**

Bellabeat could use these insights to encourage users to reduce sedentary time and increase regular daily movement. The company could use personalized reminders, activity goals, and progress notifications to motivate users to be more active. Marketing campaigns could also emphasize the benefits of tracking both daily activity and sleep to help users better understand their wellness habits.

### **What next steps would you or your stakeholders take based on your findings?**

The next step would be to test these findings using a larger and more diverse dataset collected over a longer period. Bellabeat could then use the results to develop and test features or marketing campaigns focused on increasing daily movement and reducing sedentary time. The company should monitor user engagement and activity data to evaluate whether these strategies lead to meaningful changes in user behavior.

### **Is there additional data you could use to expand on your findings?**

Yes. A larger and more diverse dataset with more users and a longer observation period would help determine whether the identified trends remain consistent across a broader population. Additional demographic data, such as age and gender, could

help identify differences between user groups. More complete sleep data would also allow for a more reliable analysis of the relationship between sleep and physical activity.

## **FINAL CONCLUSION**

The analysis shows that fitness trackers can provide useful information about users' activity and sleep patterns. However, the dataset has important limitations. It includes only 33 users and covers approximately one month, while sleep data is available for only a subset of the records. Therefore, the patterns identified in this analysis cannot be assumed to represent the wider population of fitness tracker users. A larger dataset, for example with hundreds of users and a longer observation period, would help determine whether the same trends remain consistent across a broader population.

### **Top three recommendations:**

1. Encourage regular daily movement.

Bellabeat could use personalized reminders and activity goals to encourage users to reduce sedentary time and incorporate more movement into their daily routines.

2. Use activity patterns to personalize engagement.

Since activity levels vary across the week, Bellabeat could use user activity data to provide personalized notifications and encouragement based on individual activity patterns.

3. Promote combined activity and sleep tracking.

Bellabeat could emphasize the value of monitoring both activity and sleep to give users a more complete view of their wellness habits.